

M-ary QAM Orthogonal Expanded Walsh Codes for CDMA Networks

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Abstract

An innovative M-ary QAM Orthogonal Expanded Walsh Code (OEW) suited for DS-CDMA spreading is presented in this paper. By using a novel M-ary Quadrature Amplitude Modulation (QAM), an OEW chip can be expanded from binary Walsh code to non-binary OEW form. Since of more extended coordinates of modulation constellation, it provides improved channel anti-interference and transmission security. OEW's arbitrary code length allows an arbitrary channel number, so code length can be chosen to suit the number of users for improved efficiency. This plus the higher security and anti-interference abilities make OEW of interest to designers of reliable CDMA networks.

1 Introduction

In DS-CDMA (Direct-Sequence Code Division Multiple Access) systems, both pseudo-noise (PN) sequences and orthogonal codes have been extensively used [1]. CDMA systems offer high spectrum efficiency together with other features such as multi-path resistance, inherent frequency diversity, interference rejection and the use of advanced antenna and receiver structures. As a result, CDMA technology is attracting growing interest for cellular mobile systems.

Many papers have applied fixed-length orthogonal Walsh codes in CDMA systems. The Walsh code is the most common orthogonal code used in CDMA applications. Xia *et al.* shows that Walsh functions could be divided into several cyclically orthogonal subsets, and application of cyclically orthogonal subsets for DS-CDMA was discussed [2]. Lee involves Walsh-quadrature phase-shift keying (Walsh-QPSK) pseudo-noise modulation for both coherent and non-coherent DS-CDMA systems [3]. Braithwaite presents the effects of Walsh code selection and baseband filtering on the statistical moments of band-limited forward-link CDMA signals [4]. Huang *et al.* applied complementary Walsh coding on optical CDMA coder/decoders over arrayed-waveguide grating routers [5].

Due to the secure and reliable requirements of CDMA systems, many researches and applications were involved into higher protection and reliability. Crosby and Wallach [6, 7] first exploit weaknesses in hash function implementations and provide backtracking behavior to deny the service attack. For achieving evasion through backtracking, articles in [8, 9, 10] present and show that normalization combined with stateful analysis to remove protocol ambiguities can foil evasion attempts, although it may affect stream semantics. Shankar and

Paxson [11] address semantics by providing an online database of network attributes without the risk of changing stream semantics.

Recently, a new Orthogonal Expanded Walsh Code (OEW) [12] is involved to increase channel anti-interference and security capabilities on CDMA system. It is a subset of the PN sequences but is a superset of the traditional Walsh code. Besides, it can be applied on the multi-rate CDMA system. Unlike Walsh code, which is always binary, OEW can be M-ary. Nevertheless, the innovative OEW with MPSK is suitable for single transmission manner but not for multiple transmission. In this paper, we propose an M-ary QAM OEW which can be properly working on secure DS-CDMA system since more code candidates can be supported. It allows modulating one of M signal symbols expressed within one chip of OEW, thereby allowing enhanced channel anti-interference capability without decreasing the transmission efficiency.

2 Orthogonal Expanded Walsh Codes

OEW can spread traditional binary Walsh code into non-binary form, increasing anti-interference and security because of OEW's elongated horizontal length. Whereas previous studies on Walsh codes used a $2^n \times 2^n$ Hadamard matrix, OEW uses an $m \times n$ matrix, where $m \leq n$. The OEW is obtained by first using the Gram-Schmidt Orthogonalization method [13] to find the orthogonal basis. Give a k -dimensional subspace $W \neq \{0\}$ in R^n and a basis $S = \{u_1, u_2, \dots, u_k\}$ of W . An orthogonal basis $T = \{v_1, v_2, \dots, v_k\}$ of W can be found with the following steps:

Step 1. Let $v_1 = u_1$.

Step 2. Compute vectors $v_2, v_3, \dots, v_k$ successively by

$$v_i = u_i - \frac{v_1^T u_i}{v_1^T v_1} v_1 - \frac{v_2^T u_i}{v_2^T v_2} v_2 - \dots - \frac{v_{i-1}^T u_i}{v_{i-1}^T v_{i-1}} v_{i-1}, \text{ for } 2 \leq i \leq k$$

Let $T = \{v_1, v_2, \dots, v_k\}$ be an orthogonal basis of W . Then, vector v_i is the projection of u_i onto the span $(v_1, v_2, \dots, v_{i-1})$ for $2 < i < k$ and can be represented as an OEW.

Step 3. Normalize each of the vectors $v_1, v_2, \dots, v_k$.

One can find an m-by-m orthogonal matrix by using the Gram-Schmidt method to generate binary or non-binary vectors orthogonalized to each other. Denote $\otimes$ is a Kronecker product operation [14]. The generation steps of OEW are given as following:

Step 1. Find an m by m orthogonal matrix $T_{m \times m}$ by using the Gram-Schmidt method, where

$$T_{m \times m} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mm} \end{pmatrix}.$$

Step 2. By using Kronecker product, an OEWC code $(T_{m \times m})^{\otimes i}$ can be constructed as $T_{m \times m}$, where

$$(T_{m \times m})^{\otimes i} = \underbrace{(T_{m \times m} \otimes T_{m \times m} \otimes \cdots \otimes T_{m \times m})}_{i \text{ times}}.$$

Although the above generated OEWC code is a square matrix, one can also use an m by n ($n \leq m$) orthogonal base by dispatching n rows of $T_{m \times m}$ to formulate a general form of OEWC as follows.

Theorem 1: If an orthogonal basis is $\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$, the

matrices $H_1 = H_{2^0} = 1$ and

$$H_{2^p} = \begin{pmatrix} a_{11}H_{2^{p-1}} & a_{12}H_{2^{p-1}} & \cdots & a_{1n}H_{2^{p-1}} \\ a_{21}H_{2^{p-1}} & a_{22}H_{2^{p-1}} & \cdots & a_{2n}H_{2^{p-1}} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}H_{2^{p-1}} & a_{m2}H_{2^{p-1}} & \cdots & a_{mn}H_{2^{p-1}} \end{pmatrix} \text{ is also orthogonal.}$$

Proof:

$$H_{2^p} \cdot H_{2^p}^T = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}^T =$$

$$\begin{pmatrix} \sum_{i=1}^n (a_{1i}^2) & 0 & \cdots & 0 \\ 0 & \sum_{i=1}^n (a_{2i}^2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sum_{i=1}^n (a_{mi}^2) \end{pmatrix}$$

$H_{2^p} \cdot H_{2^p}^T$ is diagonal; therefore, the row vectors of H_{2^p} are orthogonal to each other. If the row vectors of H_{2^k} are orthogonal to each other, then $H_{2^k} \cdot H_{2^k}^T$ is a diagonal matrix.

Besides, we have

$$H_{2^{k+1}} H_{2^{k+1}}^T = \begin{pmatrix} a_{11}H_{2^k} & a_{12}H_{2^k} & \cdots & a_{1n}H_{2^k} \\ a_{21}H_{2^k} & a_{22}H_{2^k} & \cdots & a_{2n}H_{2^k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}H_{2^k} & a_{m2}H_{2^k} & \cdots & a_{mn}H_{2^k} \end{pmatrix} \begin{pmatrix} a_{11}H_{2^k} & a_{12}H_{2^k} & \cdots & a_{1n}H_{2^k} \\ a_{21}H_{2^k} & a_{22}H_{2^k} & \cdots & a_{2n}H_{2^k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}H_{2^k} & a_{m2}H_{2^k} & \cdots & a_{mn}H_{2^k} \end{pmatrix}^T$$

$$= \begin{pmatrix} \sum_{i=1}^n (a_{1i}^2) \cdot H_{2^k} H_{2^k}^T & 0 & \cdots & 0 \\ 0 & \sum_{i=1}^n (a_{2i}^2) \cdot H_{2^k} H_{2^k}^T & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sum_{i=1}^n (a_{mi}^2) \cdot H_{2^k} H_{2^k}^T \end{pmatrix}$$

which is also a diagonal matrix. Hence, the row vectors of $H_{2^{k+1}}$ are orthogonal to each other. By mathematical induction, we know that the row vectors of H_{2^p} are orthogonal to each other for all positive integer p .

Lemma 1: If two orthogonal basis are A and B, the matrix $A \otimes B$ is also orthogonal.

Proof: One can combine theorem 1 and Kronecker product to get the proof.

3 M-ary QAM on OEWC

Due to that the formulated OEWCs may be non-binary, how to modulate the new arbitrary-length OEWC as transmission becomes a challenge. In this paper, we apply a novel M-ary Quadrature Amplitude Modulation (QAM) to generate a constellation about M signal units expressed within one chip of the OEWC. The QAM is a modulation scheme in which two sinusoidal carriers, one exactly 90 degrees out of phase with respect to the other, are used to transmit data over a given physical channel. Because the orthogonal carriers occupy the same frequency band and differ by a 90 degree phase shift, each can be modulated independently, transmitted over the same frequency band, and separated by demodulation at the receiver. Therefore, during OEWC transmission, the transmitted data can be spread by OEWC M-ary QAM modulation. The receiver recovers data by mapping OEWC M-ary QAM demodulation on each received signal and decodes them as Lemma 2's constellation signal mapping.

Lemma 2: A signal (a, b) can be a coordinate of constellation as a mapping value, $a+b$ on I-Q plane, where $a+b$ is expressed as an OEWC chip on QAM.

Proof: Since a and b are represent as voltage values on I-axis and Q-axis, as two signal accumulated, it just accumulate the voltage on I-axis and Q-axis. Hence, the mapping value of the signal (a, b) is just the same as $a+b$.

By induction, we can make sure that the multiple signals accumulation can work correctly as Lemma 2 proved. For instance, a 3×3 matrix of SOEWC $[\{1, 2, 2\}, \{2, 1, -2\}, \{2, -2, 1\}]$ can use 25-QAM to handle the modulation mapping on transmitter's encoder. The 25-QAM produces sixteen symbols, which are enough for the case. By Lemma 2, we can translate

non-binary value '1' in OEWC as (1, 0) symbol signal, '-1' as (0, -1) symbol signal, '2' as (0, 2) symbol signal, '-2' as (0, -2) symbol signal. For a complex case as three users transmit signals {1, 2, 2}, {2, 1, -2}, {2, -2, 1} together, the received accumulated signal is ought to be {5, 1, 1}. Since the first symbol of the received signal is '5'='1'+ '2'+ '2', by using Lemma 2, the accumulated signal is (1, 4) which is just mapped as '5' on 25-QAM constellation. Similarly, the second symbol '4'='2'+ '1'+ '1', the accumulated signal is (2, 2) which is just mapped as '4'. The third symbol '1'='2'+ '-2'+ '1', the accumulated signal is (3, -2) which is just mapped as '1'. As we select the maximum value of a and b about whole signals included the accumulated signals, we can choose a proper M-ary QAM. For the above case, the maximum value is 4, therefore, we choose the 81-ary QAM to construct the constellation mapping.

For signal and coordinate translation, it is necessary to do a proper mapping to decrease the coordinates in constellation. However, How to select a proper signal mapping is a further research and challenge. One can use a linear mapping method to match the system choice, such as in Lemma 3 using a linear equation to map the OEWC signal value and coordinate in M-ary constellation.

Lemma 3: A signal (a, b) can be a coordinate of constellation as a mapping value, $a \times 2^1 + b \times 2^0 = 2a + b$ on I-Q plane, where $2a + b$ is expressed as an OEWC chip on QAM.

Proof: Same as Lemma 2.

For CDMA network security, network intrusion detection systems may be placed at a role within the challenges of the alarming approaches lie in the adaptive modeling of group traffic. Groups are often formed in an ad-hoc, cluster, or cellular manner. The members in a group should communicate in a secure way and for efficiency reasons. Secure group communication requires scalable and efficient group membership with appropriate access control measures to protect data and to cope with potential compromises. Since the orthogonal CDMA codes are assigned in a cell or cluster or service set in a local local-area network, user or channel number is essential factor on implementation. Moreover, there may be multiple OEWCs generated and existed since of the Gram-Schmidt orthogonalization method and Kronecker product rule. Hence, system can choose multiple OEWCs for group matching on a cluster conveniently. In our scheme, system can generate a code pool mapped and copied with n_m OEWCs. Suppose there is n_m users in a chosen cluster of a CDMA system. For this cluster, security staff will choose and assign n_m OEWCs from code pool. The user number of each chosen OEWC must be greater than the user number n_m of the mapped cluster ($n_n \geq n_m$). At last, the data are transmitted block-by-block sequentially and cyclic modulated/demodulated by the chosen n_m OEWCs.

4 Performance and Results

In this section, we provide simulation result for the proposed M-ary QAM OEWC and compare it with Walsh code.

Basically, the symbol error rate of QAM system [14] can be formulated as

$$P_s \leq 4Q\left(\sqrt{\frac{3k}{M-1} \frac{E_s}{N_o}}\right)$$

where $\frac{E_s}{N_o}$ is the SNR per symbol, M is number of signal

constellations for M-ary QAM and $k=\log_2 M$. However, due to our system use multiple symbols to formulate a M-ary QAM OEWC chip, the SNR of per OEWC chip can be modified as

$$\frac{E_c}{N_o} = \frac{E_s}{N_o} L_c = \left(\frac{P_R}{P_N}\right) L_c$$

where L_c is processing gain, P_R is the average received signal power at the intended receiver, and P_N is the average noise power in signal bandwidth. For an OEWC, the processing gain L_c is the dimension of orthogonal matrix and denoted as

$$L_c = W/R$$

where W is transmitted signal bandwidth and R is original message bandwidth. Hence, we perform a simulation to compare the chip error rate of 3×3 theoretical 64-ary/16-ary QAM, 64-ary/16-ary QAM SOEWCs with 4×4 and 8×8 Walsh codes (shown in figure 1) for comparison of theoretical QAM and M-ary QAM SOEWCs with Walsh codes. Performance result shows the OEWC error rate is good for anti-MAI or secure applications. Note that the error rate of the Walsh codes are better than M-ary QAM SOEWCs since the processing gains of Walsh codes are greater than M-ary QAM SOEWCs. We also perform a simulation of different M-ary QAM SOEWCs as varying L_c (shown in figure 2) to compare theoretical chip error rate as L_c varying. The result depicts that chip error rate becomes smaller as L_c larger, i.e. anti-interference capability enhances as SOEWC L_c increases. It is because the higher L_c , the more coordinates in constellation.

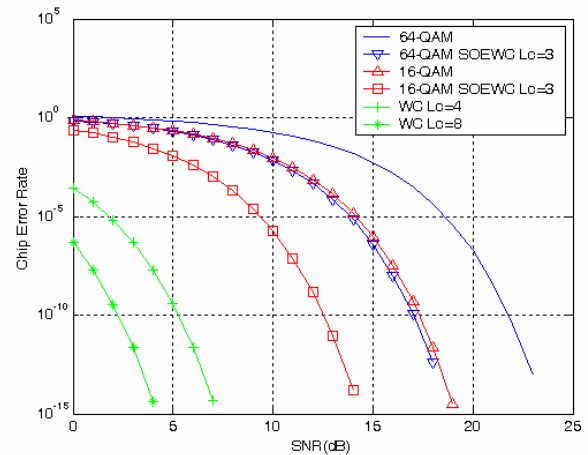


Figure 1. Chip Error Rate of 3×3 theoretical 64-ary/16-ary QAM, 64-ary/16-ary QAM SOEWCs and 4×4 and 8×8 Walsh codes

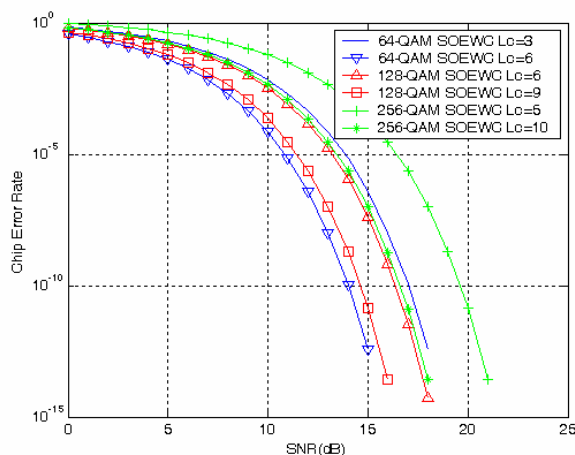


Figure 2. Chip Error Rate of different M-ary-QAM SOEWCs as L_c varying

The above analysis shows that when compared with traditional Walsh code, the M-ary SOEWC needs more processing gain for M-ary QAM. On the other hand, SOEWC's arbitrary coding length gives designers greater options in choice of code, and also provides higher security and enhanced anti-interference capabilities. These advantages make OEWC of interest to designers of the next generation of wireless or optical applications.

5 Conclusions

This paper introduces a new family of orthogonal expanded Walsh codes (OEWCs) with M-ary QAM modulation. Although requiring an increase in processing gain, due to more code candidates, OEWC allows M-ary QAM modulation of codes to provide enhanced channel anti-interference and channel security. Further, the M-ary QAM OEWC provides an arbitrary channel number whereby code length can be chosen to suit the number of users, improving efficiency. The OEWC can be of interest to designers of the reliable 3G systems and beyond CDMA networks nevertheless.

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